CRAM-ABSA: Multi-Task Rationale Alignment for Aspect-Based Hotel Review Assessment

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Abstract. Online hotel reviews help travelers choose rooms and help managers improve their services. However, existing aspect-based sentiment analysis (ABSA) systems still face three main problems: sentiment from one aspect often leaks into another, models ignore the natural order of rating levels, and predictions lack clear text evidence. We propose **CRAM-ABSA**, a two-stage framework that combines text filtering and retrieval with multi-task deep learning. In Stage 1, negation-aware NLTK rules and dense Embedding-Based Retrieval (EBR) select the most relevant sentences for each aspect. In Stage 2, a BERT model combines 20 engineered tabular features using a learnable gate, reduces aspect overlap with an orthogonal Gram-matrix penalty, and keeps rating order consistent using conditional ordinal regression (CORN). Tested on 9,990 six-aspect hotel reviews against three baseline models, CRAM-ABSA achieves strong Macro-F1 gains, very low cross-aspect leakage, fewer large rating errors, and clear text evidence.

Keywords: Aspect-Based Sentiment Analysis, Hotel Reviews, Multi-Task Learning, Rationale Alignment, Conditional Ordinal Regression, Embedding-Based Retrieval

1 Introduction

1.1 Background and Motivation

Online reviews guide travelers' booking choices [20] and help hotel managers spot service issues [21], monitor quality, and build loyalty [19,26]. Since one review often mixes opinions across different aspects (e.g., praising staff while criticizing WiFi), document-level scores are insufficient, motivating Aspect-Based Sentiment Analysis (ABSA) via Transformers [1] and hybrid models [4].

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1.2 Challenges and Knowledge Gaps

Existing hospitality ABSA systems face three main problems: (1) keyword and rule-based pipelines [4] often fail on negation (e.g., *"not bad"*); (2) standard BERT heads [1] allow negative sentiment from one aspect to leak into nearby aspects, while standard classification loss—even in graph models [9]—ignores the natural rating order (Negative < Neutral < Positive); and (3) existing baselines [1,4,9] predict labels without extracting clear text evidence for managers.

1.3 Research Questions

To compare CRAM-ABSA against three strong published baselines—BERT-Base [1], BM25-CNN [4], and WSD-BiLSTM-GCN [9]—this study answers three research questions:

- **RQ1 (Multi-Aspect Sentiment Classification):** How accurately does CRAM-ABSA classify sentiment across all six hotel aspects on an imbalanced dataset?
- **RQ2 (Leakage Control and Ordinal Consistency):** Can orthogonal masking and conditional ordinal regression reduce cross-aspect leakage and large rating errors?
- **RQ3 (Evidence Extraction):** How well do negation-aware filtering and embedding retrieval extract short, accurate text evidence for each aspect?

1.4 Key Contributions

We present **CRAM-ABSA** (*Contrastive Rationale-Aligned Multi-Task Architecture with Embedding-Based Retrieval*) with five contributions:

(i) **Two-Stage Pipeline:** Combines Stage 1 negation-aware NLTK filtering and Embedding-Based Retrieval (EBR) with Stage 2 multi-task BERT to predict six aspects (*Service, Facility, Experience, Amenity, Loyalty, Branding*) across four states (*Not Mentioned, Negative, Neutral, Positive*).

(ii) **Residual Feature Gating:** A 2-layer MLP with a learnable scalar gate initialized at $\alpha = 0.28$ injects 20 tabular features into BERT hidden vectors without distorting language representations.

(iii) **Orthogonal Rationale Masking:** A Gram-matrix orthogonality loss separates co-occurring aspects, cutting cross-aspect leakage from 54.47% (BM25-CNN) to 0.24%.

(iv) **Conditional Ordinal Regression (CORN):** Ordinal heads penalize errors by rating distance ($0 < 1 < 2$), reducing Macro-MAE to 0.3520.

(v) **Strong Benchmark Results:** On an independent 999-review test split, CRAM-ABSA leads across 6 metrics, achieving +20.00% Macro-F1 (72.55% vs. 52.55% for BERT-Base) and +20.18% to +61.07% in Rationale F1 (70.84%).

2 Related Work

2.1 Aspect-Based Sentiment Analysis in Hospitality

ABSA covers aspect extraction and sentiment classification [18]. Hotel studies evolved from classical ML and NLP [7,8] to LSTMs [29], zero-shot learning [5], Vietnamese hotel review analysis [10,25], and cross-aspect BiLSTMs with ordinal regression [30]. However, our baselines remain limited: BERT-Base [1] isolates aspects, BM25-CNN [4] uses static features that fail on negation, and WSD-BiLSTM-GCN [9] lacks ordinal distances and rationale extraction.

2.2 Rationale Alignment and Multi-Task Learning

Multi-task learning fits hospitality's multi-dimensional satisfaction [6,13,27], with recent work using TreeSHAP [16], BERT with PLS-SEM [28], SERVQUAL [17], and Outcome-Driven Innovation [12] to explain service quality. Brand studies [19,26] further show that loyalty depends on multiple touchpoints, motivating multi-task rationale alignment.

2.3 Retrieval-Augmented Text Mining

Retrieval-Augmented Generation (RAG) [23], SBERT dense matching [24], and metadata-grounded retrieval [2] reduce unsupported claims by anchoring predictions in real text. Hospitality conversational AI studies [14,22] also stress grounding outputs in customer sentences, motivating CRAM-ABSA's Embedding-Based Retrieval (EBR).

3 Proposed Methodology (CRAM-ABSA)

3.1 Problem Formulation and Pipeline Architecture

Given a customer review $R = (w_1, w_2, \ldots, w_n)$, we follow hotel analytics studies [3,11] to model six aspects: $\mathcal{A} = \{\text{Service}, \text{Facility}, \text{Experience}, \text{Amenity}, \text{Loyalty}, \text{Branding}\}$. For each aspect $a \in \mathcal{A}$, the model predicts one of four states $y_a \in \{\text{Not Mentioned}, \text{Negative}, \text{Neutral}, \text{Positive}\}$ (with ordered sentiment $y_a^{\text{sent}} \in \{0, 1, 2\}$ when mentioned) and extracts a token rationale mask $\mathbf{m}_a \in (0, 1)^n$ highlighting supporting evidence $E_a \subseteq R$.

As shown in Fig. 1, the workflow runs in five stages: (1) splitting 9,990 hotel reviews into 80% training, 10% validation, and 10% test sets, (2) engineering 20 review-level features, (3) Stage 1 negation-aware NLTK cleaning and embedding-based retrieval (EBR), (4) Stage 2 multi-task BERT modeling with orthogonal rationale masking and CORN ordinal regression, and (5) benchmarking across six evaluation metrics.



Fig. 1. Overall research pipeline of the proposed study, showing the end-to-end workflow from data extraction and 20-feature engineering to dual-stage CRAM-ABSA modeling and six-metric benchmarking.

3.2 Stage 1: Negation-Aware Linguistic Processing and Embedding Retrieval

Customer reviews often contain informal phrases and negation words that flip sentiment. Stage 1 prepares clean text signals and retrieves aspect-relevant sentences before neural modeling:

- **Negation-Aware Linguistic Processing (NLTK):** Each review R is lowercased, contractions (e.g., *"can't"*, *"n't"*) are expanded, and negation/contrast words (*"not"*, *"never"*, *"no"*, *"hardly"*, *"barely"*, *"scarcely"*, *"without"*, *"but"*) are preserved during stopword removal. Polarity is flipped when negation precedes an opinion word (e.g., *"not clean"*).
- **Embedding-Based Retrieval (EBR):** Inspired by RAG [23] and dense embeddings [24], we encode domain keywords and training spans with `bert-base-uncased` into aspect prototypes $\mathbf{p}_a \in \mathbb{R}^{768}$. Relevance between each sentence or token vector $\mathbf{e}_{s_i} \in \mathbb{R}^{768}$ and aspect a is measured by cosine similarity:

$$\mathrm{Sim}(s_i, a) = \frac{\mathbf{e}_{s_i} \cdot \mathbf{p}_a}{\|\mathbf{e}_{s_i}\| \|\mathbf{p}_a\|} \tag{1}$$

  Top-scoring spans guide downstream attention toward real aspect evidence.

3.3 Stage 2: Multi-Task CRAM-ABSA Architecture

Stage 2 combines a shared BERT encoder, residual gating for 20 tabular features, orthogonal rationale masking, and conditional ordinal regression heads (Fig. 2).

As shown in Fig. 2, Stage 2 has four connected parts:

- **Shared Contextual Backbone:** A pretrained `bert-base-uncased` encoder reads [CLS] R [SEP] to produce token states $\mathbf{H} \in \mathbb{R}^{n \times 768}$ and summary vector $\mathbf{h}_{\mathrm{CLS}} \in \mathbb{R}^{768}$.
- **Residual Feature Gating:** Twenty tabular features $\mathbf{f} \in \mathbb{R}^{20}$ (aspect counts, polarity balance, negation density, EBR similarity) pass through a two-layer MLP ($20 \to 256 \to 768$, LayerNorm, GELU, Dropout 0.15) to form $\mathbf{f}_{\mathrm{emb}} \in \mathbb{R}^{768}$, fused via learnable scalar gate α (initialized at 0.28):

$$\mathbf{H}_{\mathrm{fused}} = \mathbf{H} + \alpha\, \mathbf{f}_{\mathrm{emb}}, \quad \mathbf{h}_{\mathrm{fused}} = \mathbf{h}_{\mathrm{CLS}} + \mathbf{f}_{\mathrm{emb}} \tag{2}$$

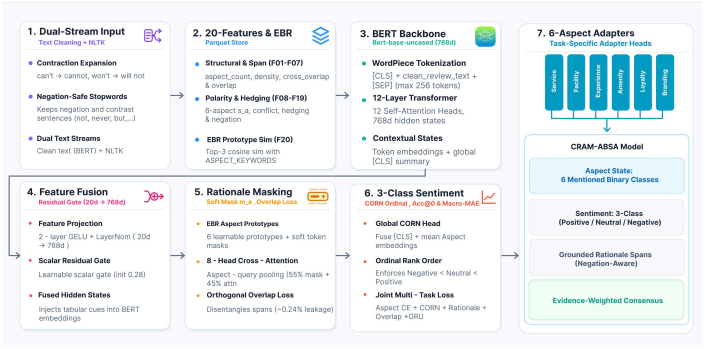


Fig. 2. Architecture of the proposed CRAM-ABSA model, combining Stage 1 negation-aware NLTK processing and EBR with Stage 2 BERT encoding, 20-feature residual gating, orthogonal rationale masking, and CORN ordinal regression.

– **Orthogonal Rationale Masking and Overlap Loss:** To prevent cross-aspect token leakage [30], EBR prototypes and 8-head cross-attention over $\mathbf{H}_{\text{fused}}$ predict six rationale masks $\mathbf{m}_a \in (0,1)^n$ trained with weighted BCE ($\mathcal{L}_{\text{rat}}$, positive weight 2.5). An overlap loss ($\mathcal{L}_{\text{overlap}}$) combines an off-diagonal Gram-matrix cosine penalty and a token-excess ReLU penalty, cutting leakage to 0.24%.
– **Aspect, CORN Ordinal, and Joint Loss Heads:** Each aspect uses a four-class classifier with Cross-Entropy and label smoothing (0.03) for $\mathcal{L}_{\text{aspect_CE}}$. A global CORN head takes $[\mathbf{h}_{\text{fused}}; \bar{\mathbf{z}}] \in \mathbb{R}^{1536}$ ($\bar{\mathbf{z}}$ is the mean of the six aspect embeddings) to predict two ordered thresholds ($y > 0$, $y > 1$) for $\mathcal{L}_{\text{CORN}}$, alongside a GRU commentary head ($\mathcal{L}_{\text{com}}$). The joint loss is:

$$\begin{aligned}\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{aspect_CE}} + \beta_{\text{CORN}}\mathcal{L}_{\text{CORN}} + \beta_{\text{rat}}\mathcal{L}_{\text{rat}} \\ + \beta_{\text{overlap}}\mathcal{L}_{\text{overlap}} + \beta_{\text{com}}\mathcal{L}_{\text{com}}\end{aligned} \tag{3}$$

with $\beta_{\text{CORN}} = 0.20$, $\beta_{\text{rat}} = 0.25$, $\beta_{\text{overlap}} = 0.35$, and $\beta_{\text{com}} = 0.05$.

4 Experimental Evaluation

4.1 Dataset and 20-Feature Engineering

We evaluate CRAM-ABSA on `TripAdvisor_EN.json`, containing 9,990 anonymized English hotel reviews collected from TripAdvisor in Vietnam in a prior unpublished project, split 80/10/10 into 7,992 training, 999 validation, and **999 test** reviews. Each review is labeled across six aspects (*Service*, *Facility*, *Experience*, *Amenity*, *Loyalty*, and *Branding*) with four states (*Not Mentioned, Negative, Neutral, Positive*).

Fig. 3 shows strong aspect imbalance (*Service* and *Facility* are common; *Loyalty* and *Branding* are rare), and Fig. 4 shows that positive sentiment dominates all six aspects. To handle this skew, we build 20 tabular features in four

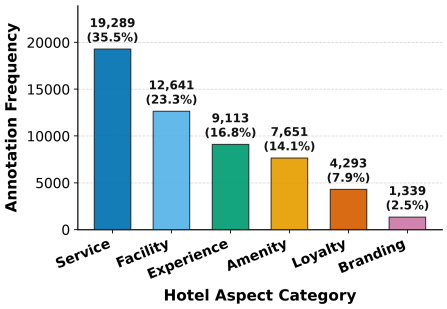


Fig. 3. Aspect frequency distribution across the 9,990 review corpus.

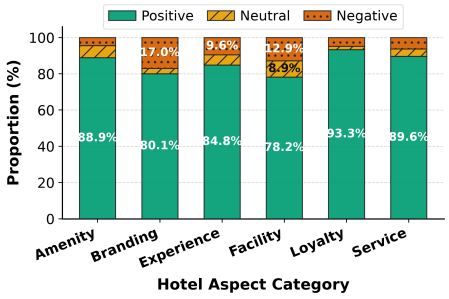


Fig. 4. Sentiment breakdown across the six evaluated hotel aspects.

groups: (1) *Aspect Structure* (Feats 01–03, 14: count, main aspect, share, coverage); (2) *Sentiment Polarity* (Feats 04, 08–13, 17: net sentiment, conflict flag, six aspect scores, polarity strength); (3) *Text Evidence* (Feats 05–07, 15–16: evidence density, mean length, overlap flag, hedging, consensus); and (4) *Linguistic and EBR* (Feats 18–20: clean token ratio, negation density, Top-3 EBR centroid similarities).

4.2 Baseline Models and Implementation Details

We compare CRAM-ABSA against three published baselines on the exact same 999-review test split:

1. **BERT-Base (Bimaputra and Sutoyo, 2023 [1]):** Fine-tuned BERT-base with one classifier head per aspect.
2. **BM25-CNN (Namee et al., 2023 [4]):** Hybrid BM25, TF-IGM, SVM, and Multi-Region CNN.
3. **WSD-BiLSTM-GCN (Nadeem et al., 2025 [9]):** Word Sense Disambiguation and Graph Convolutional Networks over syntax trees.

With SEED = 42, CRAM-ABSA trains for 8 epochs (best validation checkpoint) using AdamW (lr 3×10^{-5} for BERT, 10^{-4} for heads, 10% linear warmup, cosine decay, per-GPU batch 16 with 2 accumulation steps). Baselines follow fixed schedules: BERT-Base (3 epochs, AdamW, lr 2×10^{-5}), BM25-CNN (8 epochs, Adam, lr 10^{-3}, batch 64), and WSD-BiLSTM-GCN (6 epochs, AdamW, lr $3 \times 10^{-5}/10^{-3}$).

4.3 Evaluation Metrics

We evaluate all models on six metrics: (1) **Acc@0 (%) ↑:** exact-match sentiment accuracy (zero rating error); (2) **6-Aspect Macro-F1 (%) ↑:** average Macro-F1 across all six aspects and four states; (3) **Macro-MAE ↓:** ordinal distance error (penalizing opposite-polarity errors by 2); (4) **Rationale F1 (%) ↑:** token-level overlap F1 with human evidence spans; (5) **Leakage (%) ↓:** share of tokens assigned to two or more aspects; and (6) **Test Loss ↓:** total multi-task test loss.

4.4 Overall Quantitative Benchmark Results

Table 1 and Fig. 5 show the results of CRAM-ABSA and the three baseline models on the 999-review test set.

Table 1. Benchmark evaluation results on the independent test split (999 samples). Arrows indicate whether higher (↑) or lower (↓) values are preferred. Best results across each column are shown in **bold**.

Model Architecture	Test Loss ↓	Acc@0 (%) ↑	Macro-F1 (%) ↑	Macro-MAE ↓	Rat-F1 (%) ↑	Leak (%) ↓
CRAM-ABSA (Ours)	**0.2552**	92.39	**72.55**	**0.3520**	**70.84**	0.24
BERT-Base [1]	0.5143	92.29	52.55	0.4823	50.66	**0.02**
BM25-CNN [4]	0.6032	91.69	44.22	0.6743	32.65	54.47
WSD-BiLSTM-GCN [9]	0.5227	**92.59**	52.12	0.4756	9.77	0.22

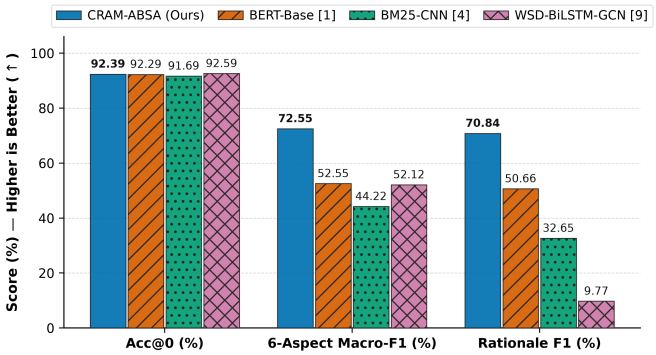


Fig. 5. Quality metric comparison on the 999-sample independent test split across the four evaluated models, covering Acc@0, 6-Aspect Macro-F1, and Rationale F1 (higher ↑ is better).

CRAM-ABSA leads in 4 of 6 metrics: (1) *Test Loss (0.2552):* >50% lower than baselines (0.5143–0.6032); (2) *6-Aspect Macro-F1 (72.55%):* +20.00 points over BERT-Base (52.55%), +20.43 over WSD-BiLSTM-GCN (52.12%), and +28.33 over BM25-CNN (44.22%); (3) *Macro-MAE (0.3520):* below WSD-BiLSTM-GCN (0.4756), BERT-Base (0.4823), and BM25-CNN (0.6743); (4) *Rationale F1 (70.84%):* ahead of BERT-Base (50.66%), BM25-CNN (32.65%), and WSD-BiLSTM-GCN (9.77%); (5) *Leakage (0.24%) and Acc@0 (92.39%):* close to WSD-BiLSTM-GCN (92.59%) while avoiding BM25-CNN's 54.47% leakage (BERT-Base is 0.02%).

4.5 Metric Interpretation Across Research Questions (RQ1–RQ3)

We now explain what these benchmark numbers mean for our three Research Questions:

- **RQ1 (Multi-Aspect Sentiment Classification):** Baselines reach high Acc@0 (91.69–92.59%) mostly by predicting common *Not Mentioned* and positive labels (Fig. 3, Fig. 4), but their Macro-F1 drops to 44.22–52.55% on rare aspects (*Loyalty*, *Branding*) and negative/neutral cases. Combining text embeddings with the 20 tabular features via residual gating lifts CRAM-ABSA's Macro-F1 to 72.55% at 92.39% Acc@0.
- **RQ2 (Leakage Control and Ordinal Consistency):** CORN ordinal regression lowers Macro-MAE to 0.3520 (vs. 0.4756–0.6743) by reducing opposite-polarity errors. Meanwhile, Fig. 6 shows ground-truth spans rarely overlap (8 cases total). BM25-CNN's keyword rules cause 54.47% leakage, and BERT-Base's 0.02% leakage comes from separate heads with lower Rationale F1 (50.66%). Orthogonal masking keeps CRAM-ABSA's leakage at 0.24% with 70.84% Rationale F1.
- **RQ3 (Evidence Extraction):** Fig. 7 shows human evidence spans are short (medians 39–66 characters). Baselines score lower (9.77–50.66% Rationale F1) due to missing span alignment or negation errors, whereas Stage 1 negation filtering and EBR help CRAM-ABSA reach 70.84%.

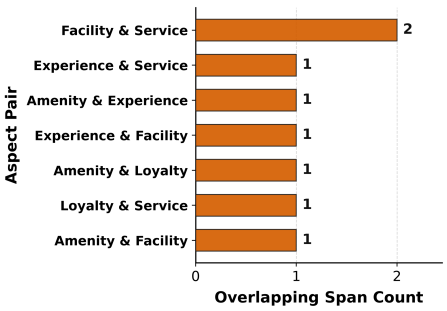


Fig. 6. Counts of overlapping text coordinate spans across hotel aspect pairs.

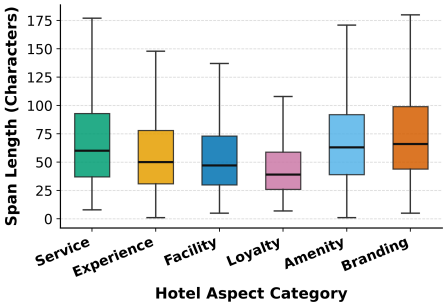


Fig. 7. Distribution of rationale span lengths (in characters) across the six aspects.

5 Discussion and Conclusion

5.1 Practical and Managerial Implications

Our results offer clear value for hospitality research and hotel management. For research, CRAM-ABSA resolves key trade-offs among aspect coverage, leakage,

and ordinal accuracy while making predictions explainable through text evidence [16]. For management, tracking all six aspects (*Service, Facility, Experience, Amenity, Loyalty, Branding*) gives deeper insight than a single rating: operators can apply penalty-reward analysis [13] to spot root causes of complaints and use extracted spans for fast service recovery [15].

5.2 Conclusion, Limitations and Future Work

We presented **CRAM-ABSA**, a multi-task model for six-aspect hotel review assessment and evidence extraction that combines NLTK filtering, dense EBR, orthogonal masking, 20-feature residual gating, and CORN regression to overcome baseline limits [1,4,9]. On the 999-sample test split, CRAM-ABSA leads in 4 of 6 metrics: lowest Test Loss (0.2552), highest 6-Aspect Macro-F1 (72.55%, +20% gain), lowest Macro-MAE (0.3520), and highest Rationale F1 (70.84%), with 0.24% leakage and 92.39% Acc@0. A limitation is our single-split evaluation; future work will add multi-fold validation, parameter ablation grids, and multilingual transfer across Southeast Asian hotel markets.

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– **Code availability:** Available in the CRAM-ABSA-EBR-Hotel-Reviews repository (https://github.com/trietnguyen2611/CRAM-ABSA-EBR-Hotel-Reviews).
– **Author contribution:**
  • **Triet Minh Pham Nguyen:** Designed the method and drafted the manuscript.
  • **Tien Hoang Duy Nguyen:** Performed data cleaning, EDA, and benchmark experiments.
  • **Duy Bui Anh Nguyen:** Performed data cleaning, EDA, and benchmark experiments.
  • **Thu Le:** Supervised the study, guided methodology, and revised the manuscript.
  • **Huong Bui:** Supervised the study, guided methodology, and revised the manuscript.
  • **Hanh Thi My Le:** Provided and curated the hotel review dataset, and reviewed the manuscript.

  All authors read and approved the final manuscript.

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